

Michael Chou

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EDUCATION

Tufts University

Bachelor of Science in Electrical Engineering, Minor in Embedded Systems: GPA: 3.95

Medford, MA

May 2027

Relevant Courses: Embedded Systems, Computer Organization, Parallel Computing, Digital IC Design, Communication Systems, Digital Electronics, Modern Physics, Data Science, Networked Embedded Systems, Computer Vision, Linear Algebra, Probability Theory, Linear Systems, Circuits and Electronics

EXPERIENCE

Tufts University Department of Electrical and Computer Engineering

Digital Logic Teaching Assistant

Medford, MA

August 2025 - Present

- Restructured VHDL-based course to SystemVerilog by transitioning all lab assignments to Verilog and adopting APiO as the FPGA development environment
- Lead 2 weekly lab sections and an office hours guiding students through hands-on circuit building and combinational and sequential logic FPGA design projects to 80+ students

Massachusetts Institute of Technology Koch Institute

Research Assistant

Cambridge, MA

August 2025 - Present

- Developing a nanoparticle drug loading device for exosome encapsulation using acoustofluidic techniques
- Validating and optimizing loading efficiency by tuning parameters like particle size and acoustic power, with quantification via imaging flow cytometry

Tufts Engineers Without Borders

Vice President — Greenhouse Tech Lead

Medford, MA

October 2023 - Present

- Vice President of a 200+ member club overseeing 3 community-driven engineering projects in Malawi, Ecuador, and Massachusetts; coordinating multidisciplinary engineering teams and partnerships, securing funding, and planning international travel for assessment and implementation of sustainable water, sanitation, and community infrastructure solutions
- Led a 10+ member team to design a Wi-Fi-enabled temperature and humidity monitoring system for the campus greenhouse; currently transitioning the prototype to a custom ESP32-C3 PCB by removing high-power peripherals, and implementing deep-sleep duty cycling to minimize power usage and enable long-term, low-maintenance field deployment
- Built an on-campus greenhouse, designing and implementing a fully automatic watering system; all vegetables donated to the SHC Food Pantry

University of Pennsylvania — Iot4ag REU funded by the NSF

Researcher: Optical Sensor Detector Systems and Robotics

Philadelphia, PA

May 2025 - August 2025

- Developed an automated hyperspectral imaging pipeline in Python with Ximea API and embedded system integration with on-device machine learning on the NVIDIA Jetson Xavier NX for The Beast, a self-driving farming robot, enabling real-time analysis of optical metasurface-based colorimetric leaf sensors
- Proposed and led a project correcting hyperspectral angle-detection errors, developing a calibration model that expanded measurement capability from 0° to $\pm 40^\circ$ with 1.3° error
- Trained in semiconductor fabrication processes – including photolithography, thin film deposition, and etching – at the Quattrone Nanofabrication Facility in the Singh Center for Nanotechnology

Foresight AI (For Waymo)

3D Gaussian Splatting Intern

Mountain View, CA

June 2024 - August 2024

- Initiated real world implementation and tested 3D Gaussian Splatting (3DGS) for automated 3D world reconstruction as a replacement for the 3D mesh reconstruction pipeline for training Waymo cars
- Generated metrics on car detection via YOLOV8 between 3DGS rendered images and original images and other 3D world reconstruction methods, like NERF, demonstrating improved detection accuracy

University of Pennsylvania Physics Department

Electronics Engineer Intern

Philadelphia, PA

April 2023 - July 2023

- Built and tested PCBs which will be used to amplify signals from Cherenkov light with photomultiplier tubes to study neutrino interactions under the supervision of Mitch Newcomer and Professor Hugh Williams

PROJECTS

5 Stage Pipelined ARM CPU

- Developing a multi-cycle pipelined 64 bit ARM-LEGv8 CPU with branch forwarding and hazard detection in VHDL
- Perform rigorous hardware verification through GHDL simulation and GTKWave signal analysis

Guitar Hero Game on FPGA

- Developed a real-time "Guitar Hero" inspired game on an FPGA using VHDL
- Implemented VGA display output, a scoring system, and real-time audio synthesis based on guitar controller inputs

Hand Signal Instrument

- Created a gesture-controlled musical instrument that uses a webcam to recognize hand signals in real time with a convolutional neural network (CNN)
- The trained model runs on a laptop which sends the classification through UART to an STM32 microcontroller (CMSIS framework), which interprets the classification input and plays corresponding notes through a speaker based on the detected gesture

SKILLS

Programming: VHDL, SystemVerilog, Embedded C, Python, C++, Linux, Git, ARMv8 Assembly, Makefiles

Tools and Frameworks: Radiant, APiO, PlatformIO, KiCad, Onshape, LTSpice, Waveforms, Oscilloscopes, ARM Cortex-M, Arduino, CMSIS, ESP-IDF